

Vol 5 — Quantum Mechanics from D_IV⁵ (Pedagogical Bridge)

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

BST Physics Curriculum | Casey Koons + team (Lyra, Elie, Grace, Keeper, Cal A. Brate)

Status: SCAFFOLD v0.1 (Saturday 2026-05-23 Wave 2) — chapter content TO BUILD Lead: Lyra (primary, pedagogical bridge from substrate to standard QM) + Elie (computational verification)

Reader target

Undergraduate + early graduate physics. Reads Vol 0 (Substrate Foundation) as prerequisite; recovers standard QM (Griffiths / Shankar / Sakurai level) as the substrate-level emergent limit at scales above substrate-tick. Pedagogical bridge volume — connects BST substrate framework to physics-community-standard QM curriculum.

Prerequisites

- Vol 0 Substrate Foundation (D_IV⁵ + five integers)
- Vol 1 Ch 2 + Ch 6 + Ch 7 Substrate Hilbert Space + Operator Zoo + Dynamics (basic prerequisite)
- Standard undergraduate QM background (helpful, not required)

BST integer focus

Vol 5 is the substrate $\rightarrow$ standard QM emergence pedagogical bridge. Demonstrates: - Standard Hilbert space $L^2(\mathbb{R}^3)$ emerges as macroscopic limit of Bergman $H^2(D_IV^5)$ at coarse-grained scale - Position + momentum + angular momentum + spin standard QM operators emerge from substrate-coset-Cartan-decomposition operators (Vol 1 Ch 6 6/6 zoo) - Schrödinger equation $i\hbar \partial|\psi\rangle/\partial t = H|\psi\rangle$ emerges from substrate dynamics $H_{\text{sub}} = \text{Casimir on } L^2(D_IV^5; L_\lambda)$ (Vol 1 Ch 7) at macroscopic scale - Born rule $|\psi|^2 = \text{Bergman reproducing-kernel evaluation (K67 RATIFIED Tuesday)}$ - Measurement formalism + POVMs derive from substrate observer-bandwidth-bounded recording (T2469 SCMP framework) [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This

sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

Chapter outline (10-12 chapters, TO BUILD)

Ch	Title	Reader takeaway	Cross-link
1	From Substrate to Standard Hilbert Space	$L^2(\mathbb{R}^3)$ emerges from Bergman $H^2(D_{IV}^5)$ at macroscopic limit	Vol 1 Ch 2
2	Position + Momentum + Heisenberg	Standard QM Heisenberg algebra emerges from coset Cartan decomposition	Vol 0 Ch 4 §4.2 v0.5; Vol 1 Ch 6
3	Angular Momentum + Spin	Standard $SO(3)$ angular momentum + half-integer spin from substrate $SO(5) \times SO(2)$ reduction	Vol 1 Ch 6 + Paper #133 spin-statistics
4	Schrödinger Equation	Standard time-dependent + independent Schrödinger from substrate H_{sub} dynamics	Vol 1 Ch 7 dynamics
5	Heisenberg Picture + Path Integral	Standard Heisenberg + Feynman path integral as substrate dynamics representations	Vol 1 Ch 7 + Ch 9
6	Hydrogen Atom + Atomic Spectra	Standard hydrogen energy levels + selection rules from substrate-natural Casimir spectrum	Vol 1 Ch 5 Casimir

Ch	Title	Reader takeaway	Cross-link
7	Born Rule + Measure- ment	Born = Bergman (K67 RATIFIED); POVMs + projective measurement	T2469 SCMP framework [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893); the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S =$ 2.82759 ± 0.00051 , 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]
8	Bell-CHSH + Quantum Correlations	Standard Bell experiments + Tsirelson bound + the BST sub-Tsirelson 1/8 falsifier — FIRED (Poh et al. 2015, 41.9σ ; K1893, register E4, 2026-09-11)	T2399 + Calibration #17; E4
9	Identical Particles + Spin- Statistics	Pauli exclusion + boson/fermion partition from Pin(2) Z_2 grading	Paper #133 v0.2

Ch	Title	Reader takeaway	Cross-link
10	Decoherence + Classical Limit	Macroscopic classical emergence from substrate ensemble-marginalization	T2469 SCMP + Vol 4 Ch 3 boundary [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]
11	POVMs + Quantum Information Basics	Born = Bergman extension to general POVMs	SP-31 #283
12	Pedagogical Bridge Synthesis	Standard QM curriculum recovery + cross-curriculum reading guide	Vol 1 Ch 2-9 + Vol 0 Ch 4-7 cross-link

Approximately 35% existing BST coverage (Vol 1 + Vol 0 substrate framework provides scaffolding); 65% to build at pedagogical-bridge accessibility.

Cross-volume reference map

Inputs from Vol 0: - D_{IV}^5 + five integers + Bergman $H^2(D_{IV}^5)$ - Operator zoo 12/14 STRUCTURALLY VERIFIED (T2419, T2422, T2425, T2421, T2399, H_sub Elie + T2470/T2471/T2472) [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.] - Substrate-coset Cartan decomposition + $\text{Pin}(2)$ Z_2 grading

Inputs from Vol 1: - Bergman Hilbert space + Casimir algebra (Ch 2 + Ch 5) - Schrödinger / Heisenberg / path integral framework (Ch 7) - Operator zoo 6/6

framework-complete (Ch 6) - Substrate-tick UV-completeness (Ch 10)

Outputs to: - Vol 7 Information Theory (Bell-CHSH + Born rule + POVMs) - Vol 8
Biology (substrate-coupled observer framework) - Vol 9 Condensed Matter (many-
body QM emergent from substrate)

— Vol 5 INDEX v0.1, Saturday 2026-05-23 Wave 2 scaffold

2026-09-11 status of every chapter (added 2026-09-13, K1895)

Derived-core shelf; legacy chapters. The Spine (../Spine_DIV5_QM_GR_SM/) is current where they disagree; the registry wins over both.

chapter	status
Curriculum_Vol5_Architectural_Scaffold	scan-clean; May tiers
Curriculum_Vol5_Ch10_Decoherence_Classical_Limit	scan-clean; May tiers
Curriculum_Vol5_Ch11_POVMs_Quantum_Scale_Fundamentals	scan-clean; May tiers
Curriculum_Vol5_Ch12_Pedagogical_Bridge_Synthesis	scan-clean; May tiers
Curriculum_Vol5_Ch1_From_Substrate_to_Structure	scan-clean; May tiers
Curriculum_Vol5_Ch2_Position_Moments	scan-clean; May tiers
Curriculum_Vol5_Ch3_Angular_Moments	scan-clean; May tiers
Curriculum_Vol5_Ch4_Schrodinger_Equation	scan-clean; May tiers
Curriculum_Vol5_Ch5_Heisenberg_Picture	scan-clean; May tiers
Curriculum_Vol5_Ch6_Hydrogen_Atom	scan-clean; May tiers
Curriculum_Vol5_Ch7_Born_Rule_Measurement	scan-clean; May tiers
Curriculum_Vol5_Ch8_Bell_CHSH_Quantum_Relations	scan-clean; May tiers
Curriculum_Vol5_Ch9_Identical_Particles	scan-clean; May tiers